

Additional details

Software: Qucs (Quite Universal Circuit Simulator)

Supported OS: Windows, Mac, Ubuntu

Size: Windows - 54MB, Mac - 106MB, Ubuntu - 247.3MB

Latest stable release: 0.0.17 on June 23, 2013

The dialogue is closed by pressing the Enter key or pressing the Ok button. Now the complete schematic for the voltage divider is ready and can be saved. This can be done by choosing the File → Save menu entry, clicking the Single Disk icon or by pressing the Ctrl+S shortcut.

The final DC voltage divider is shown in Fig. 8.

Issuing a simulation. The schematic can now be simulated. This is started by choosing the Simulation → Simulate menu entry, clicking the simulation button (the gearwheel) or by pressing the F2 shortcut. After the simulation has been done, the related data display is shown. Also, the Components tab changes its category to 'diagrams.'

Placing diagrams. Choose the tabular (list of values) diagram and place it on the data display page. By double clicking 'divided.V', the graph (values in a tabular plot) is added to the diagram. Besides the node voltage 'divided.V' the current through the DC voltage source V1.I is available. Only items listed in the dataset list can be put in the graph.

Available dataset items

Depending on the type of simulation the user performed you find the following types of items in the dataset:

```
# node.v - DC voltage at node node
# name.I - DC current through
component name
# node.v - AC voltage at node node
# name.i - AC current through
component name
# node.vn - AC noise voltage at node
node
# name.in - AC noise current through
component name
# node.Vt - Transient voltage at node
```

Operating system support for Qucs

Qucs is currently developed on GNU/Linux systems using the GNU auto tools (and CMake experimentally). For this reason it is very likely that Qucs runs on all kinds of Unix systems supporting an appropriate system interface and an installed Qt environment. It has been verified that Qucs can be installed on GNU/Linux, Solaris 2.9, MacOS 10.5, NetBSD, FreeBSD and Cygwin (Unix emulation layer for Windows). Qucs is also available natively on x86 Windows (XP/Vista/7/8) systems and experimentally on x86_64 versions of these.

Some other simulation tools

1. EasyEDA, online schematic capture, ngspice simulation and PCB design
2. Falstad Circuit Simulator Applet, Paul Falstad's online java circuit simulator
3. GeckoCIRCUITS GPL software, online applet version (Java)
4. GNU Circuit Analysis Package (GnuCap)
5. Ngspice, including Online Spice

What users have to say about Qucs

"Qucs is a very powerful circuit simulator program and has features I have yet to understand. It is also very user friendly. It is a great piece of software for anyone wanting to learn electronics and the price is right. Zero."

"This is probably the most useful open source circuit simulator I ever found. It is accurate and (once you learn a bit) really easy to use."

"Very nice product. I am a DSP guy, so I do not use circuit simulators often. I have some experience with Spice and OrCAD, but I have a very strong preference for Qucs. Easier to use and richer in features. For example, the basic 'OpAmp' is just what you need when you want to make some examples in class without drowning in the complexities of a 'real' OpAmp."

"Very nice program which I use mostly to simulate and display S-parameters files and circuits."

—Courtesy: <http://sourceforge.net/projects/qucs/>

```
node
# name.It - Transient current through
component name
# S[1,1] - S-parameter value
```

Please note that all voltages and currents are peak values and all noise voltages are RMS values at 1Hz bandwidth.

Depending on the type of graph, you have various options to choose for the graph. For a tabular graph, there is the number precision as well as type of number notation (important for complex values). Press the Ok button to close the dialogue.

In the tabular graph, as seen in <image 9>, the value of the node voltage 'divided.V' is 0.5V. That is expected since the resistors are of equal values and the DC voltage source produces 1V.

Congratulations! You made your

first successful simulation using Qucs.

License

Qucs is a free software released under the General Public License (GPL). It comes with complete source code. Every user of the program is allowed and called upon (on a voluntary basis of course) to modify it for their applications, as long as the changes are made public.

The screenshots show the main schematic and data display area of the Qucs GUI on the lower right corner, the menus and icons at the upper part and the project/contents/component/action/diagram selection on the left-hand side. ●

The author is a senior technical correspondent at EFY. This article has inputs from Qucs' recently released documentation